GRADE 1 · INTRO DEBUGGING

Spot the Wrong Block

Compare the plan to the code

Name: .... Date: ....

★ **I can find the block with the wrong number.**

Plan vs code

Bit's PLAN says what each step should be. The CODE is what is really there. One block in the code has the WRONG number, so Bit misses the target. Compare the plan to the code to find the wrong block.

Bit's number path

0	1	2	3	4	5

Plan A (start at 1)

The plan is: start at 1, forward 2, forward 1. That lands Bit on 4 (the target).

Code A — has a bug

start at 1

forward 2

forward 3

1 Run Code A

Run Code A from 1. What number does Bit land on? Did it reach the target (4)?

2 Spot the wrong block

Compare Code A to the plan. Circle the block that is wrong, and write what it should be.

Plan B (start at 5)

The plan is: start at 5, back 2, back 1. That lands Bit on 2 (the target).

Code B — has a bug

start at 5

back 2

back 3

3 You try — spot it in Code B

Run Code B from 5. Where does Bit land? Compare to the plan, circle the wrong block, and write what it should be.

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Big idea

To find a bug, compare what the code DOES to what it SHOULD do. The block that does not match the plan is the wrong one.

TEACHER GUIDE · GRADE 1 · INTRO DEBUGGING

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Quick facilitation notes & answer key

What this worksheet teaches

Students find a bug by comparing the plan (what each step should be) to the program (what is there) and circling the block with the wrong number. Reading a program against what it should do is how you find where it goes wrong. No coding experience needed.

Before you start (no computer needed)

No computer — paper and a pencil. You read each prompt aloud. Optional: tape a number line 0 to 5 on the floor and mark the target, so students can walk the buggy program and feel Bit miss.

How to lead it (about 20 minutes)

1. Show them (you do). Run the buggy program — it misses the target — then compare it to the plan to find the odd block.
2. Do one together. Exercise 1 — Code A lands on 5, not 4.
3. Let them try. Students circle the wrong block and write what it should be (Ex 2), then do Code B (Ex 3).

Answer key (these are the verified correct answers)

Exercise 1 — Code A: 1, forward 2 → 3, forward 3 → stops at 5. Bit lands on 5, not the target 4.

Exercise 2 — the wrong block is the last one, forward 3; it should be forward 1 (then 1 → 3 → 4).

Exercise 3 — Code B: 5, back 2 → 3, back 3 → stops at 0. The wrong block is back 3; it should be back 1 (then 5 → 3 → 2).

If students get stuck — and ways to adjust

Watch for: circling a block without checking it against the plan. Compare each block to what it should be.

Make it easier: run the program one block at a time and stop where it goes wrong.

Make it harder: ask them to explain how they knew that block, not another, was wrong.

Ontario curriculum link (official wording)

C3.1 — solve problems and create computational representations of mathematical situations by writing and executing code, including code that involves sequential events.

In plain terms: students follow (or write) step-by-step instructions, in order, to get a result.

C3.2 — read and alter existing code, including code that involves sequential events, and describe how changes to the code affect the outcomes.

In plain terms: students read a program, change a block to fix it, and explain what the change did.

You'll know it worked when

The child circles the wrong block in each program and writes the correct block.